



Tecnológico de Monterrey

Reporte:
Actividad 4: VPN

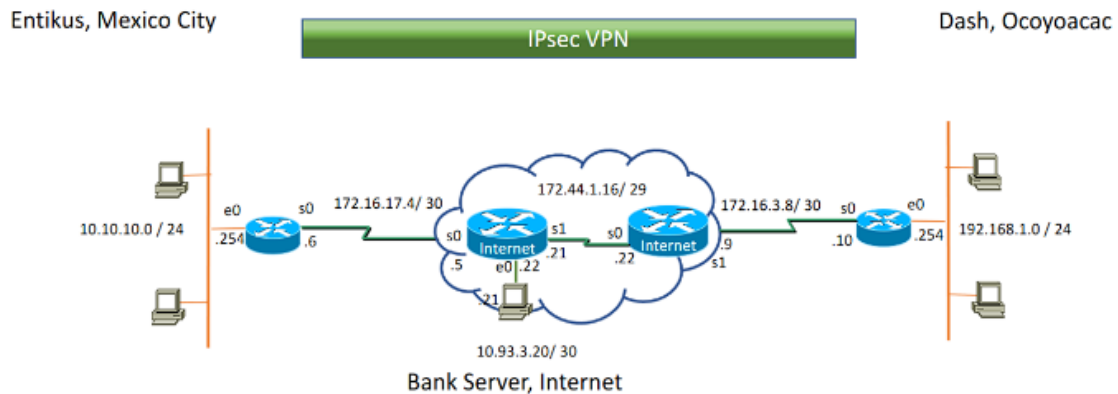
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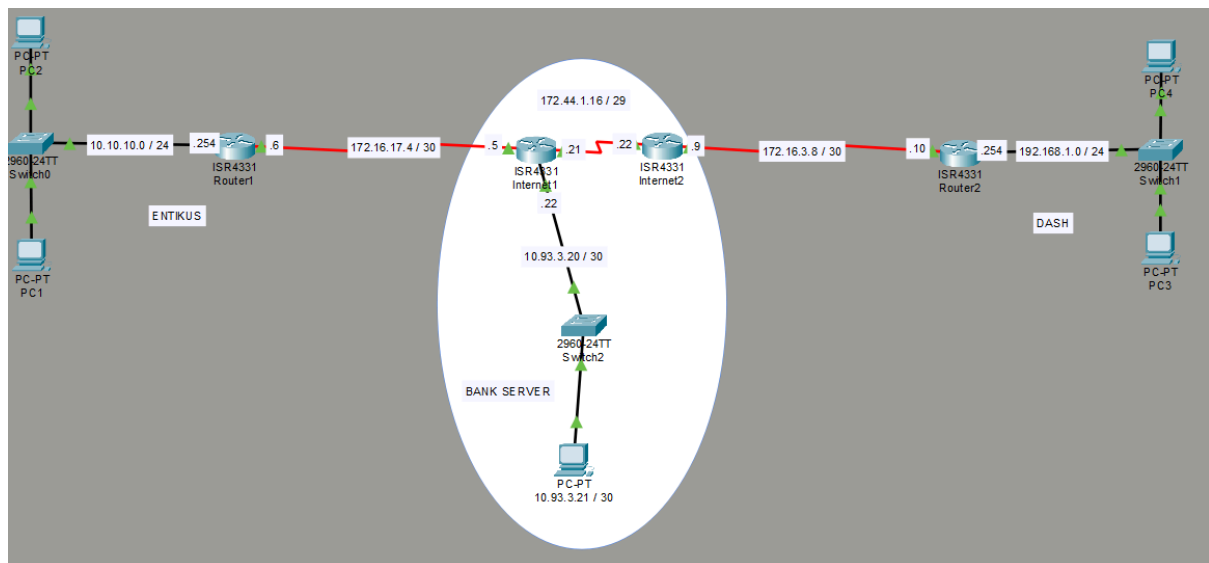
Profesor:
Jorge Rodríguez Ruiz

Instrucciones:

Entikus es una empresa que acaba de adquirir a Dash y quiere que la infraestructura desplegada de las dos empresas se conecte entre sí. Necesitas generar el siguiente escenario utilizando NAT, DHCP para las computadoras internas y una VPN entre las dos infraestructuras para que estas se puedan conectar entre sí.



Tras haber configurado la topología de las redes:



Primero configurar el servicio de DHCP para Entikus y Dash individualmente.

ENTIKUS:

None

```
Router1(config)#ip dhcp excluded-address 10.10.10.254
Router1(config)#ip dhcp pool Entikus
Router1(dhcp-config)#network 10.10.10.0 255.255.255.0
Router1(dhcp-config)#default-router 10.10.10.254
```

```
Router1(dhcp-config)#exit
Router1(config)#exit
Router1#write memory
```

DASH:

```
None
Router1(config)#ip dhcp excluded-address 192.168.1.254
Router1(config)#ip dhcp pool Dash
Router1(dhcp-config)#network 192.168.1.0 255.255.255.0
Router1(dhcp-config)#default-router 192.168.1.254
Router1(dhcp-config)#exit
Router1(config)#exit
Router1#write memory
```

Ahora configurar el servicio de NAT igual para ambas empresas.

ENTIKUS:

```
None
Router1(config)#interface gigabitEthernet 0/0/0
Router1(config-if)#ip nat inside
Router1(config-if)#exit
Router1(config)#interface serial 0/1/0
Router1(config-if)#ip nat outside
Router1(config-if)#exit
Router1(config)# access-list 101 deny ip 10.10.10.0
0.0.0.255 192.168.1.0 0.0.0.255
access-list 101 permit ip 10.10.10.0 0.0.0.255 any
Router1(config)# ip nat inside source list 101 interface
serial 0/1/0 overload
```

DASH:

```
None
Router2(config)#interface gigabitEthernet 0/0/0
Router2(config-if)#ip nat inside
```

```
Router2(config-if)#exit
Router2(config)#interface serial 0/1/0
Router2(config-if)#ip nat outside
Router2(config-if)#exit
Router2(config)# access-list 101 deny ip 192.168.1.0
0.0.0.255 10.10.10.0 0.0.0.255
Router2(config)#access-list 101 permit ip 192.168.1.0
0.0.0.255 any
Router2(config)#ip nat inside source list 101 interface
serial 0/1/0 overload
```

Por último, configurar el servicio de VPN.

ENTIKUS:

```
None
Router1(config)# crypto isakmp policy 10
Router1(config-isakmp)# encryption aes
Router1(config-isakmp)# authentication pre-share
Router1(config-isakmp)# group 2
Router1(config-isakmp)# exit

Router1(config)# crypto isakmp key 0 Clave123 address
172.16.3.10
Router1(config)# crypto ipsec transform-set VPN_SET esp-aes
esp-sha-hmac
Router1(config)# access-list 100 permit ip 10.10.10.0
0.0.0.255 192.168.1.0 0.0.0.255

Router1(config)# crypto map MAPA_VPN 10 ipsec-isakmp
Router1(config-crypto-map)# set peer 172.16.3.10
Router1(config-crypto-map)# set transform-set VPN_SET
Router1(config-crypto-map)# match address 100
Router1(config-crypto-map)# interface serial 0/1/0
Router1(config-if)# crypto map MAPA_VPN
```

DASH:

None

! Fase 1

```
Router2(config)# crypto isakmp policy 10
```

```
Router2(config-isakmp)# encryption aes
```

```
Router2(config-isakmp)# authentication pre-share
```

```
Router2(config-isakmp)# group 2
```

```
Router2(config-isakmp)# exit
```

```
Router2(config)# crypto isakmp key Clave123 address  
172.16.17.6
```

```
Router2(config)# crypto ipsec transform-set VPN_SET esp-aes  
esp-sha-hmac
```

```
Router2(config)# access-list 100 permit ip 192.168.1.0  
0.0.0.255 10.10.10.0 0.0.0.255
```

```
Router2(config)# crypto map MAPA_VPN 10 ipsec-isakmp
```

```
Router2(config-crypto-map)# set peer 172.16.17.6
```

```
Router2(config-crypto-map)# set transform-set VPN_SET
```

```
Router2(config-crypto-map)# match address 100
```

```
Router2(config-crypto-map)# exit
```

```
Router2(config)# interface serial 0/1/0
```

```
Router2(config-if)# crypto map MAPA_VPN
```

Configuraciones finales:

ENTIKUS:

None

```
Current configuration : 1444 bytes
```

!

```
version 16.6.4
```

```
no service timestamps log datetime msec
```

```
no service timestamps debug datetime msec
```

```
no service password-encryption
!
hostname Router1
!
!
!
!
ip dhcp excluded-address 10.10.10.254
!
ip dhcp pool Entikus
network 10.10.10.0 255.255.255.0
default-router 10.10.10.254
!
!
!
no ip cef
no ipv6 cef
!
!
!
!
crypto isakmp policy 10
encr aes
authentication pre-share
group 2
!
crypto isakmp key Clave123 address 172.16.3.10
!
!
!
crypto ipsec transform-set VPN_SET esp-aes esp-sha-hmac
!
crypto map MAPA_VPN 10 ipsec-isakmp
set peer 172.16.3.10
set transform-set VPN_SET
match address 100
```

```
!  
!  
!  
!  
!  
!  
!  
!  
spanning-tree mode pvst  
!  
!  
!  
!  
!  
!  
interface GigabitEthernet0/0/0  
ip address 10.10.10.254 255.255.255.0  
ip nat inside  
duplex auto  
speed auto  
!  
interface GigabitEthernet0/0/1  
no ip address  
duplex auto  
speed auto  
!  
interface GigabitEthernet0/0/2  
no ip address  
duplex auto  
speed auto  
!  
interface Serial0/1/0  
ip address 172.16.17.6 255.255.255.252  
ip nat outside  
crypto map MAPA_VPN  
!
```

```
interface Serial0/1/1
no ip address
clock rate 2000000
!
interface Vlan1
no ip address
!
ip classless
ip route 0.0.0.0 0.0.0.0 172.16.17.5
!
ip flow-export version 9
!
!
access-list 100 permit ip 10.10.10.0 0.0.0.255 192.168.1.0
0.0.0.255
access-list 101 deny ip 10.10.10.0 0.0.0.255 192.168.1.0
0.0.0.255
access-list 101 permit ip 10.10.10.0 0.0.0.255 any
!
!
!
!
!
line con 0
!
line aux 0
!
line vty 0 4
login
!
!
!
end
```

Internet1:

None

Current configuration : 835 bytes

!

version 16.6.4

no service timestamps log datetime msec

no service timestamps debug datetime msec

no service password-encryption

!

hostname Internet1

!

!

!

!

!

!

!

!

no ip cef

no ipv6 cef

!

!

!

!

!

!

!

!

!

!

!

!

!

!

spanning-tree mode pvst

!

!

!

```
!  
!  
!  
interface GigabitEthernet0/0/0  
ip address 10.93.3.22 255.255.255.252  
duplex auto  
speed auto  
!  
interface GigabitEthernet0/0/1  
no ip address  
duplex auto  
speed auto  
!  
interface GigabitEthernet0/0/2  
no ip address  
duplex auto  
speed auto  
!  
interface Serial0/1/0  
ip address 172.16.17.5 255.255.255.252  
clock rate 2000000  
!  
interface Serial0/1/1  
ip address 172.44.1.21 255.255.255.248  
!  
interface Vlan1  
no ip address  
!  
ip classless  
ip route 10.10.10.0 255.255.255.0 172.16.17.6  
!  
ip flow-export version 9  
!  
!  
!  
!
```

```
!  
!  
!  
line con 0  
!  
line aux 0  
!  
line vty 0 4  
login  
!  
!  
!  
end
```

Internet2:

```
None
Current configuration : 900 bytes
!
version 16.6.4
no service timestamps log datetime msec
no service timestamps debug datetime msec
no service password-encryption
!
hostname Internet2
!
!
!
!
!
!
!
!
!
no ip cef
no ipv6 cef
!
```

```
!  
!  
!  
!  
!  
!  
!  
!  
!  
!  
!  
!  
!  
spanning-tree mode pvst  
!  
!  
!  
!  
!  
!  
interface GigabitEthernet0/0/0  
no ip address  
duplex auto  
speed auto  
shutdown  
!  
interface GigabitEthernet0/0/1  
no ip address  
duplex auto  
speed auto  
shutdown  
!  
interface GigabitEthernet0/0/2  
no ip address  
duplex auto  
speed auto
```

```
shutdown
!
interface Serial0/1/0
ip address 172.44.1.22 255.255.255.248
clock rate 2000000
!
interface Serial0/1/1
ip address 172.16.3.9 255.255.255.252
!
interface Vlan1
no ip address
shutdown
!
ip classless
ip route 192.168.1.0 255.255.255.0 172.16.3.10
ip route 10.93.3.20 255.255.255.252 172.44.1.21
!
ip flow-export version 9
!
!
!
!
!
!
!
line con 0
!
line aux 0
!
line vty 0 4
login
!
!
!
end
```

DASH:

None

Current configuration : 1566 bytes

!

version 16.6.4

no service timestamps log datetime msec

no service timestamps debug datetime msec

no service password-encryption

!

hostname Router2

!

!

!

!

ip dhcp excluded-address 192.168.1.254

!

ip dhcp pool Dash

network 192.168.1.0 255.255.255.0

default-router 192.168.1.254

!

!

!

no ip cef

no ipv6 cef

!

!

!

!

crypto isakmp policy 10

encr aes

authentication pre-share

group 2

!

crypto isakmp key Clave123 address 172.16.17.6

!

!

```
!  
crypto ipsec transform-set VPN_SET esp-aes esp-sha-hmac  
!  
crypto map MAPA_VPN 10 ipsec-isakmp  
set peer 172.16.17.6  
set transform-set VPN_SET  
match address 100  
!  
!  
!  
!  
!  
!  
!  
!  
spanning-tree mode pvst  
!  
!  
!  
!  
!  
!  
interface GigabitEthernet0/0/0  
ip address 192.168.1.254 255.255.255.0  
ip nat inside  
duplex auto  
speed auto  
!  
interface GigabitEthernet0/0/1  
no ip address  
duplex auto  
speed auto  
shutdown  
!  
interface GigabitEthernet0/0/2  
no ip address
```

```
duplex auto
speed auto
shutdown
!
interface Serial0/1/0
ip address 172.16.3.10 255.255.255.252
ip nat outside
clock rate 2000000
crypto map MAPA_VPN
!
interface Serial0/1/1
no ip address
clock rate 2000000
shutdown
!
interface Vlan1
no ip address
shutdown
!
ip nat inside source list 101 interface Serial0/1/0
overload
ip classless
ip route 0.0.0.0 0.0.0.0 172.16.3.9
!
ip flow-export version 9
!
!
access-list 100 permit ip 192.168.1.0 0.0.0.255 10.10.10.0
0.0.0.255
access-list 101 deny ip 192.168.1.0 0.0.0.255 10.10.10.0
0.0.0.255
access-list 101 permit ip 192.168.1.0 0.0.0.255 any
!
!
!
!
```



```
!  
line con 0  
!  
line aux 0  
!  
line vty 0 4  
login  
!  
!  
!  
end
```